NLP COURSE | FACULTY OF ENGINEERING, BAR-ILAN UNIVERSITY

ElectiveMatch-BIU

Reasoned Elective Course Recommendations from Academic Transcripts and Focused Constraints

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SLIDE 1 · MOTIVATING USE CASE

The Problem: Selecting Elective Courses

Background

Engineering students at Bar-Ilan University must select elective courses aligned with specialized tracks, prerequisites, and degree tracks. The selection requires personalized trade-offs across prior grades, weekly workloads, and career goals.

Why It Matters

Elective choices directly impact cumulative GPA, eligibility thresholds for graduate school, and industry competitiveness. An ill-fitted decision frequently leads to academic overload, lower grades, and degree completion delays.

Why It Is Hard

Official syllabi are frequently outdated or lack practical details, while student forums offer highly subjective and polarized opinions. Crucial data regarding true weekly workload, exam rigor, and student fit remains fragmented and unstandardized.

How It Is Solved Today

Students rely on WhatsApp chat rumors or prompt generic LLMs that lack department-specific curriculum constraints. Even when fed a syllabus, commercial models hallucinate requirements and fail to navigate delicate GPA-workload trade-offs.

SLIDE 2 · PROJECT TASK DESCRIPTION

Task Definition, Inputs & Outputs

Formal Definition

Given an academic transcript and 3 explicit student constraints, rank the optimal elective courses and generate transparent justifications. The system balances rigid prerequisite validation with nuanced trade-off decisions against student goals.

Novelty

Moving beyond simple keyword similarity toward an expert decision model trained on localized faculty trade-offs in Hebrew. It combines curated course feature cards with localized logic while maintaining complete transcript data privacy.

System Input

Academic transcript (completed courses, cumulative GPA, specialized tracks) paired with 3 preferences: target workload (1-5), grading preference (exam vs. project), and goal (M.Sc. vs. industry). Catalog cards serve as context.

System Output

A ranked recommendation of top-3 elective courses tailored to the profile. Each course includes an evidence-backed rationale explaining the trade-offs, alongside explicit warning flags for borderline prerequisites or risks.

SLIDE 3 · MODELS AND METHODS

System Architecture & Model Strategy

1. Course Cards (Offline)

Constructing a structured database of BIU electives covering prerequisites, workload intensity, evaluation formats, and M.Sc. suitability. This serves as the consistent domain knowledge base across all models.

2. Deterministic Filter

A rule-based program logic verifying prerequisites, filtering previously completed courses, and checking track eligibility. This guarantees zero hard-constraint violations before language model scoring.

3. Ranking Model (Online)

A language model that evaluates remaining eligible courses against student preferences to output ranked, justified recommendations. It resolves conflicting constraints like GPA protection versus research ambitions.

Fine-Tuning Strategy

Fine-tuning an open-weight Hebrew-capable model (DictaBERT / LLaMA-3) using QLoRA over thousands of synthesized student cases. The model learns expert trade-off logic without sending private transcripts off-premise.

Baseline Comparison

Benchmarking against state-of-the-art zero-shot and few-shot commercial LLMs (GPT-4o) provided with the exact same input context. This tests whether general LLMs can reliably balance academic dilemmas without specialized tuning.

SLIDE 4 · DATA SPECIFICATION AND GENERATION

Data Collection, Dataset & Synthesis

Data Requirements

The project requires diverse representation across academic tracks, broad GPA distributions, and diverse student preference profiles. Training demands thousands of labeled examples, while evaluation requires verified real-world engineering transcripts.

Dataset Composition

Structured attribute cards for 25–30 Bar-Ilan engineering electives detailing requirements and workload norms. In addition, 30–50 fully anonymized genuine student transcripts collected from peers will serve as our independent Test Set.

Labeling Protocol

Expert human labeling by the team for every transcript in the Test Set, assigning ground-truth course selections and justifications. A formalized decision rubric ensures consistent judgments to establish reliable evaluation standards.

Synthetic Data Generation

Sampling randomized student vectors (track, historical grades, and constraint profiles) to synthesize transcripts via Attribute-Conditioned Generation. Rule-simulated expert labels generate ~2,000 rigorous pairs solely for model training.

SLIDE 5 · METRICS AND EVALUATION

Evaluation Metrics & Protocol

How We Measure

All models (fine-tuned vs. commercial baselines) will be evaluated on the identical real-world transcript test set. We assess performance both on hard prerequisite compliance and the quality of complex academic trade-offs.

Ground Truth

Curated expert human judgments on the 30–50 real transcripts, accompanied by documented reasoning protocols. Synthetic data remains strictly quarantined to the training set to prevent data contamination and circular bias.

Quantitative Metrics

Constraint Violation Rate (unmet prerequisites or completed courses), ranking metrics (nDCG@3 and Precision@3), and explanation quality (1-5 scale via LLM-as-a-Judge alongside human expert spot checks).

Evaluation Protocol

Evaluating across three tiers of student profiles: straightforward constraints, borderline GPA limits, and conflicting ambitions. GPT-4o is tested with standard prompts versus few-shot trade-off instructions to benchmark tuning value.